

WHAT BENDS THE CURVE?

Paris at ten: long trends, a new acceleration, and the national transformations that carry the signal

Structural breaks · Kaya pathways · five decades of national decarbonization

204 jurisdictions 97 eligible national breaks 7 mechanism families 9 constructive decarbonizations

THE QUESTION

Paris is working — but through what?

The good news is real.

Worst-case emissions scenarios have moved off the table, and global CO2 growth has slowed from +2.1% per year (1990–2015) to +0.4% per year since Paris.

The open question:

Is the global improvement a direct effect of the agreement — or the continuation of trends rooted decades earlier, which Paris coordinates and reinforces?

Our approach:

Rather than assuming treaty years are turning points, we let the data tell us when trends actually changed — globally and for every country with enough annual data — and then use historical evidence to ask why.

Why intensity comes first

Emissions can be split into three moving parts:

CO2 growth = economic growth + change in fuel mix + change in energy efficiency

Emissions cannot bend until the fuel mix or efficiency improves faster. Intensity is the leading indicator: any policy effect must show up there first — before it can ever show up in emissions.

Unless intensity improves faster than the economy grows, emissions cannot stabilize.

Finding the bend: two straight lines and a kink

Three steps, no assumptions about treaty years

1. Plot each series on a log scale, so a straight line means a steady percentage change per year.
2. Fit two straight lines that meet at a kink — the 'breakpoint'. Try every possible kink year and keep the one that fits the data best. That year is the estimate of when the trend changed.
3. Stress-test the date: re-estimate it on many statistically reshuffled versions of the series. If the estimated year barely moves, the break is credible; if it jumps around, we say so.

A breakpoint tells us when a trend changed — never, by itself, why. Historical evidence answers that separately.

(Formally: $y(t) = a + b \cdot t + d \cdot \max(t-k, 0)$; slope b before year k , $b+d$ after. Uncertainty via moving-block bootstrap; autocorrelation-robust standard errors for fixed-date tests.)

Two doors for any improvement

Carbon intensity — CO₂ per dollar of output — can only improve through one of two doors:

Cleaner energy (C/E): less CO₂ per unit of energy — fuel switching, nuclear, renewables, the power system.

Less energy per dollar (E/GDP): efficiency, and changes in what the economy makes — industry versus services.

We estimate the bend in the overall series and in both doors, for the world and for every country, with one identical procedure. Which door moved tells us what kind of transition happened — a fuel-mix bend points to energy policy or markets; an efficiency bend often points to economic change.

Which breaks do we trust — and when does a treaty get credit?

A country's break counts only if:

- the series is long enough (40+ years, with at least 10 on each side of the break);
- the two-line model fits clearly better than a single straight line;
- the estimated year stays put when the data are reshuffled; and
- the change in slope is big enough to matter, not a statistical whisker.

204 jurisdictions → 158 with estimable series → 97 countries with a break that passes every test. Breaks close together in time are read as one national transition: 107 episodes in all.

A treaty year gets credit only if all five hold:

1. the slope genuinely changes at the treaty year (by a robust statistical test);
2. the treaty year beats nearly all nearby 'placebo' years — because when a trend curves gently, almost any year looks significant;
3. the best-fitting year is within three years of the treaty;
4. the data-chosen break agrees with it; and
5. there is a documented domestic policy story that fits.

This ladder is deliberately hard to climb. Many series pass step 1; few pass them all. We hold Paris itself to the same standard.

Four acts

I

Is the global bend Paris?

Global curves bent in the 1970s; since ~2013 they are bending faster. The acceleration is real — its attribution is not yet identified.

II

What bends national curves?

97 eligible breaks scattered across five decades and seven historically validated mechanism families.

III

Did the bends count?

Transition quality separates genuine decarbonization from recessions, disruption, offshoring, and growth that outruns efficiency.

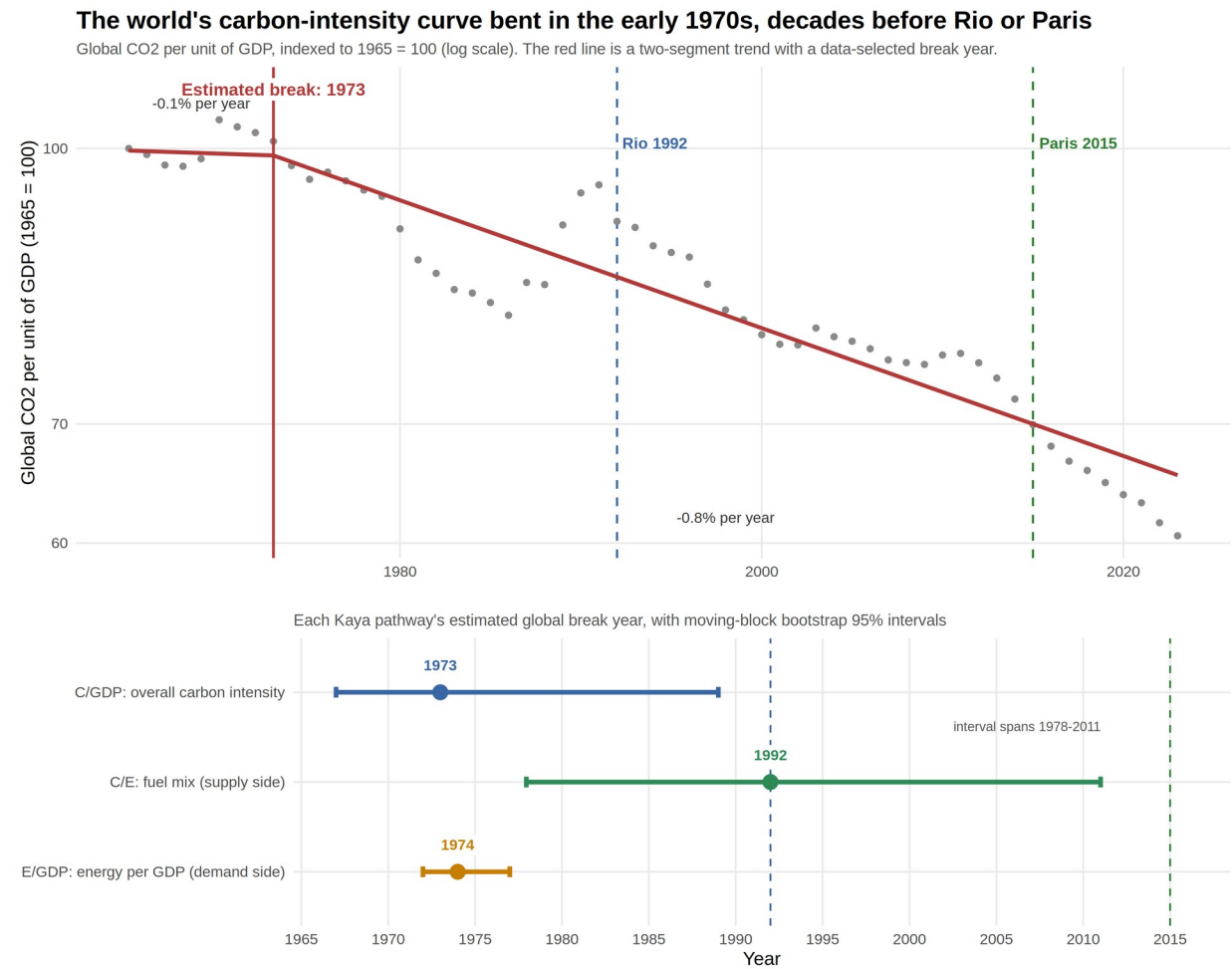
IV

Can we trust the story?

Event-timing tests against chance, external ratings, consumption-based accounting — and unresolved cases reported as such.

Global curves first bent in the 1970s

How to read it: Gray dots are the world's CO2 per dollar of output each year (1965 = 100). The red line is the best-fitting pair of straight lines; where they meet — 1973 — is the estimated break. Rio and Paris are marked for comparison: both come long after the bend. The lower panel repeats the estimate for each component; the whiskers show how uncertain each break year is.

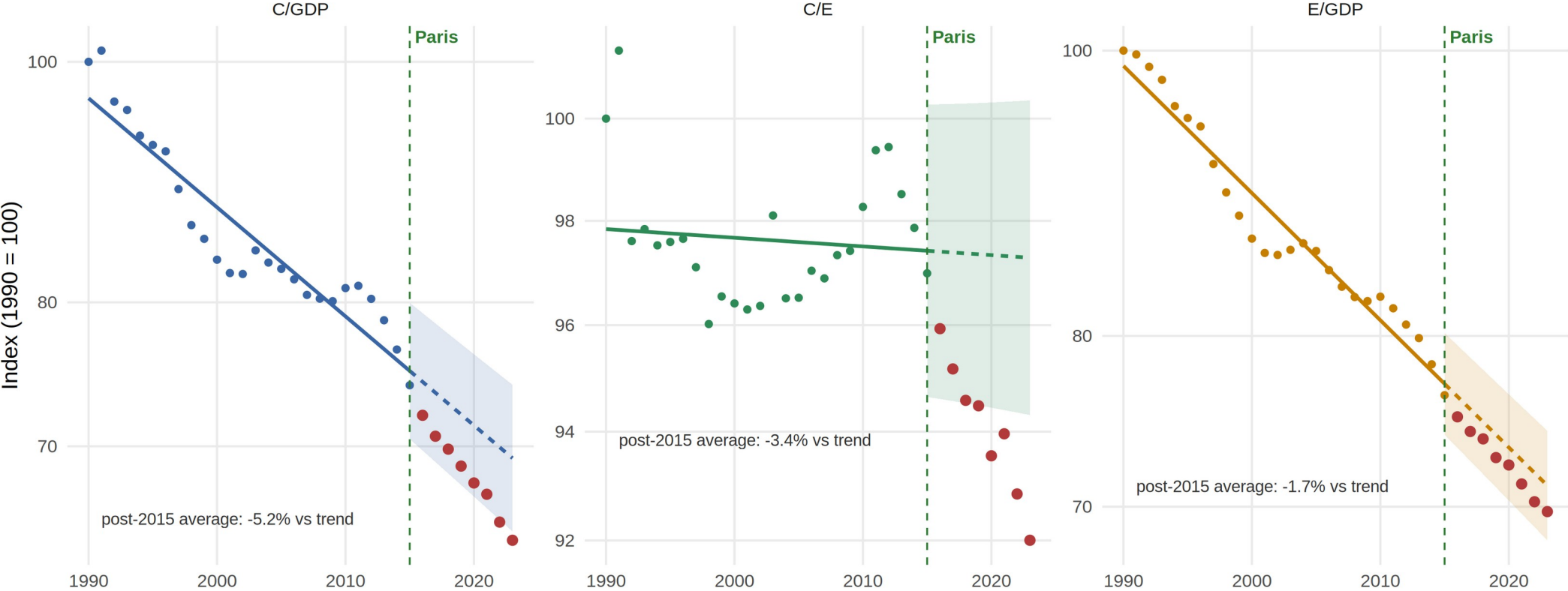


Paris at ten: ahead of trend — but the bend began earlier

How to read it: We draw the 1990–2015 trend, then extend it past Paris (dashed line) with a band showing where future years should fall if nothing changed. Red dots are the actual years since Paris. Dots below the band mean the world is now improving faster than its old trend — clearly so for the fuel mix, not at all for energy efficiency.

Since Paris, global intensity is finally outrunning its old trend - but the bend began before the treaty

Each panel: 1990 = 100 (log scale). Solid line: 1990–2015 trend; dashed: its extrapolation with a 95% prediction band; red points: observed after Paris. The fall below the band is a real supply-side acceleration (C/E -0.7 pp/yr, Newey-West t = 4.3; C/GDP -0.9, t = 2.9), but candidate onsets from 2011 to 2016 fit the full record about equally well (best: 2013), and the demand side (E/GDP -0.3, t = 1.5) shows no acceleration - so a Paris effect cannot yet be separated from the renewables cost decline that preceded it.

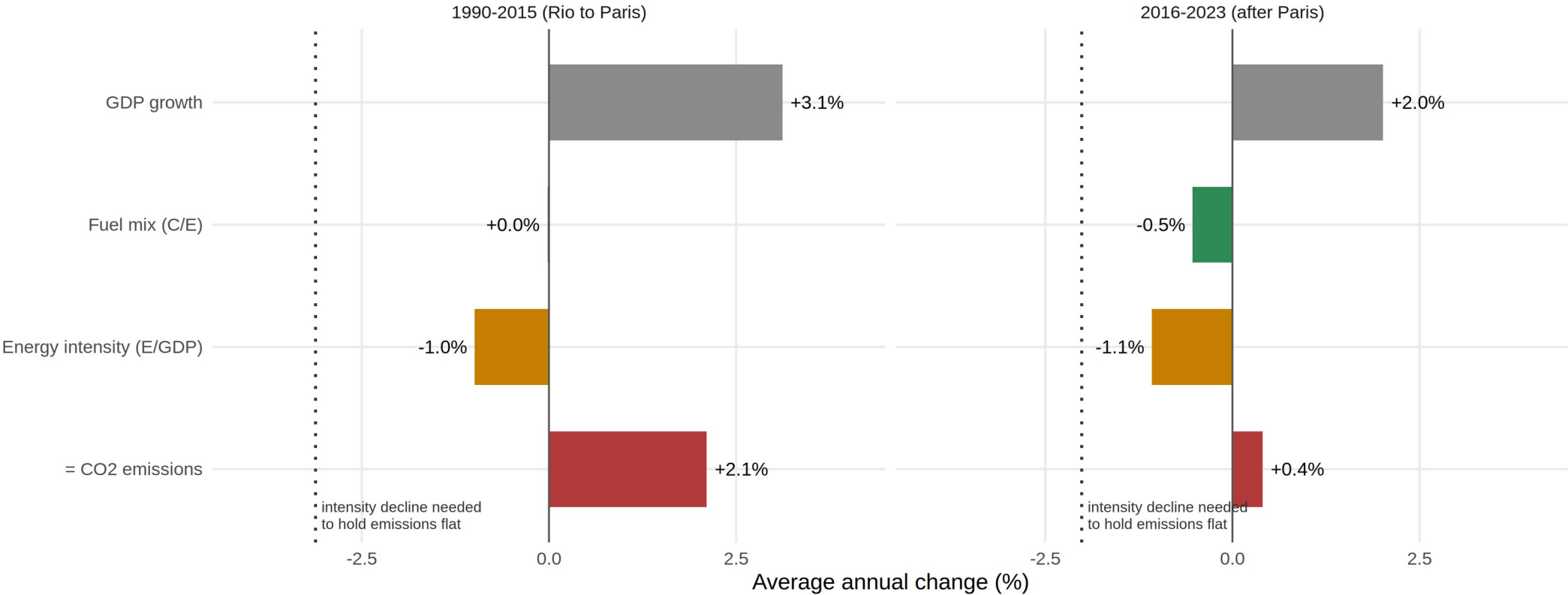


The arithmetic: intensity must fall as fast as GDP grows

How to read it: Each bar is an average annual growth rate. By construction, the red emissions bar equals the gray GDP bar plus the two colored intensity bars. Emissions stop growing only when the colored bars together are as long as the gray one — the dotted line marks that target. We are closer than before Paris, but not there.

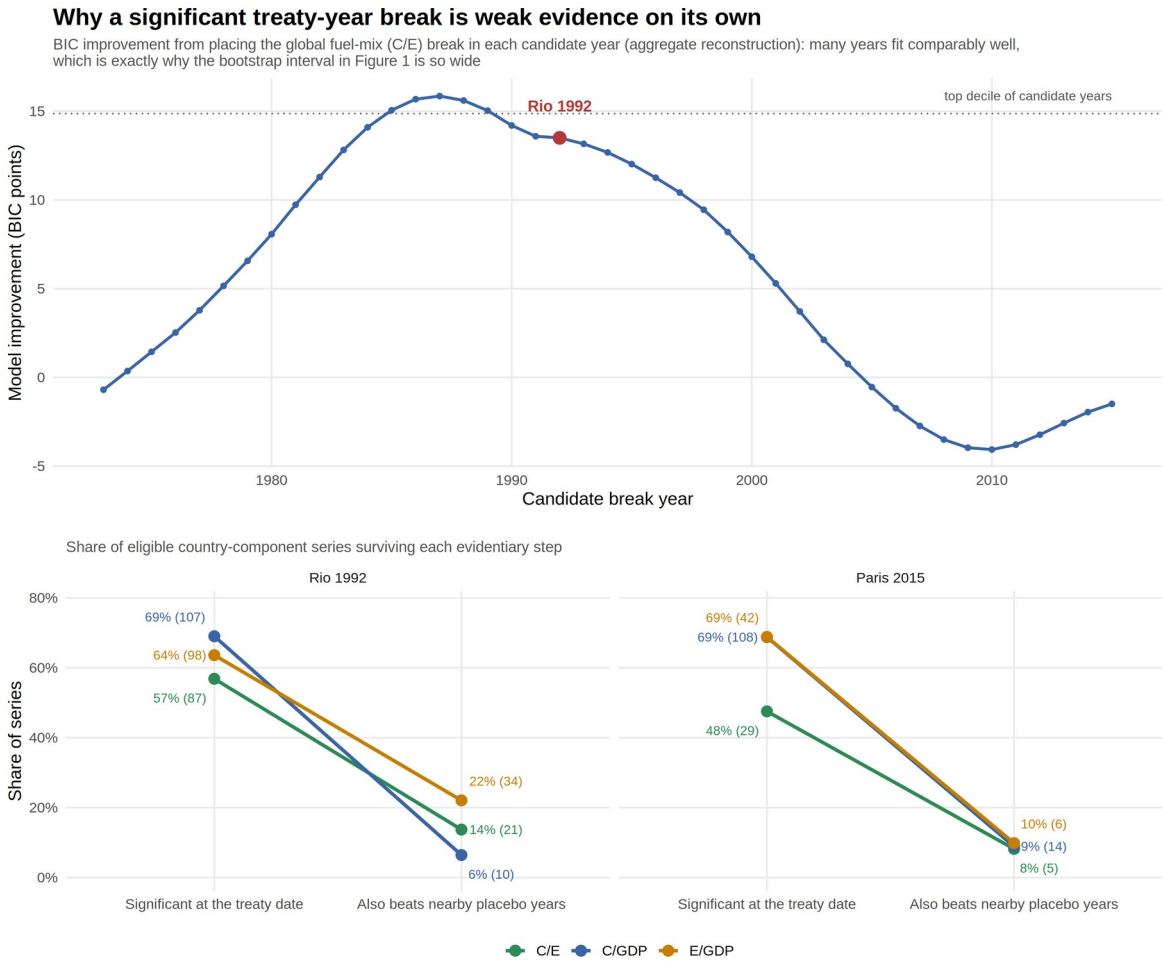
The arithmetic of bending the curve: intensity must fall as fast as GDP grows

Average annual log growth rates of the global Kaya terms. CO2 growth = GDP growth + fuel-mix change + energy-intensity change, so the emissions curve cannot bend until the intensity slopes accelerate - which is where any policy signal must appear first. Emissions growth slowed from +2.1% to +0.4% per year after 2015: roughly two-thirds of that slowdown is slower GDP growth, one-third a cleaner fuel mix.



Why a treaty-year result needs a placebo test

How to read it: Top: we place the break at every possible year, one at a time, and record how much better the model fits. Rio fits well — but so do many neighboring years, which is why a good fit at a treaty year proves little on its own. Bottom: the share of country series with a significant change at each treaty year (left point) versus the share where the treaty year also beats its neighbors (right point).



The global record can show an acceleration — not yet its author

What the data establishes

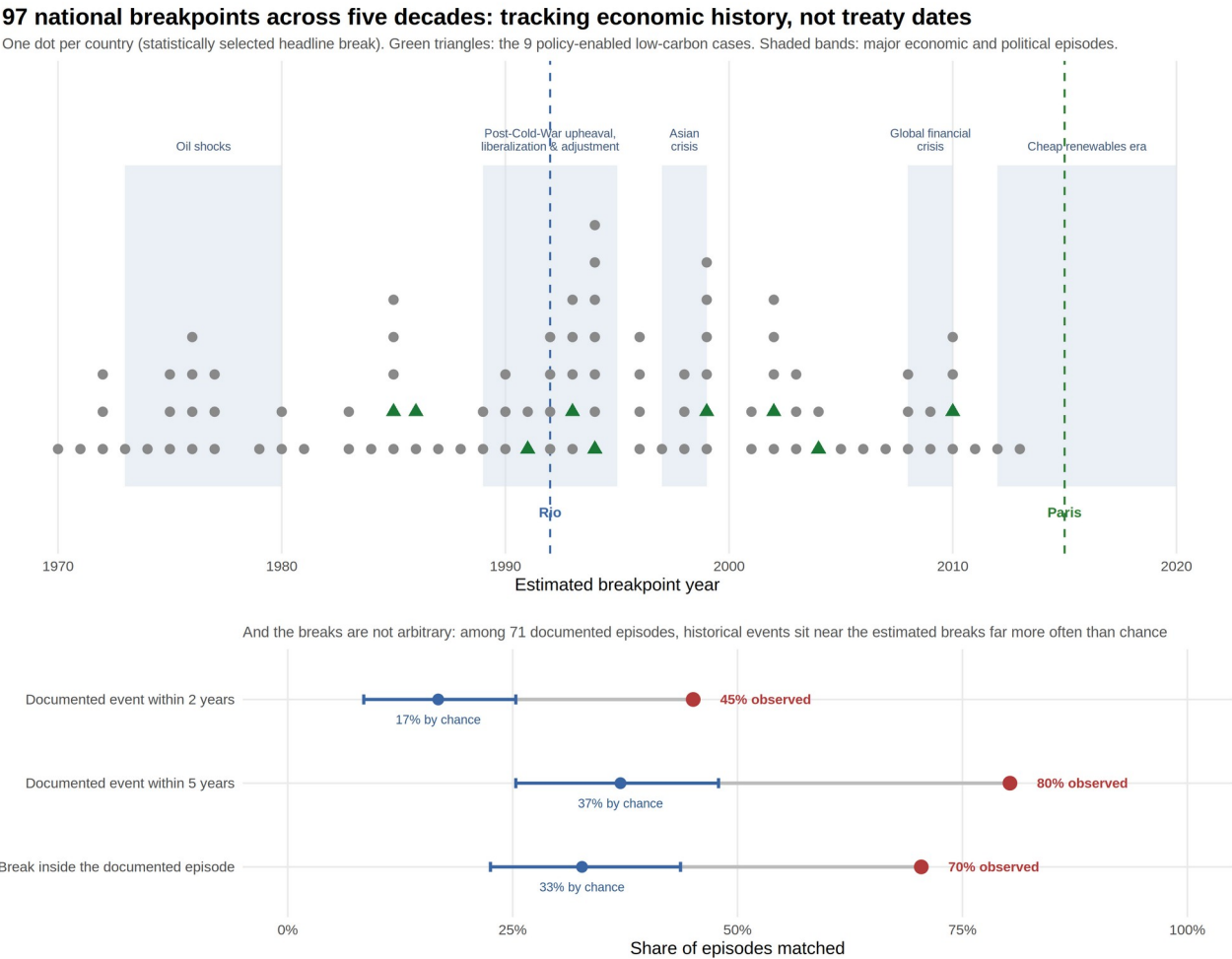
- A real post-2013 supply-side acceleration: the fuel mix is improving $-0.5\%/yr$ after a flat quarter-century; overall intensity is falling ~ 0.9 pp/yr faster than its Rio-to-Paris trend.
- Emissions growth slowed from $+2.1$ to $+0.4\%/yr$ — the direction an effective Paris requires.
- No acceleration on the demand side: energy efficiency continues its 50-year trend, no faster.

Why attribution stays open

- Timing: start years from 2011 to 2016 fit about equally well (best: 2013) — indistinguishable from the renewables cost collapse that preceded Paris.
- Power: with only 8 post-Paris years, we could only detect a very large acceleration — Paris would have had to double a 50-year trend to prove itself already.
- Aggregation: China's WTO-era coal boom flattened the global fuel-mix baseline; the world series is a China-weighted average that hides national signals.
- Design: Paris works through national pledges and national transformation — its signature should appear country by country. So we look there.

97 national breaks, five decades of economic history

How to read it: Top: each dot is one country, placed at its estimated break year; green triangles are the nine policy-enabled cases. Shaded bands mark major economic and political episodes; Rio and Paris are the two dashed lines. Bottom: how often a documented historical event falls near a country's break (red) versus what randomly chosen years would produce (blue).

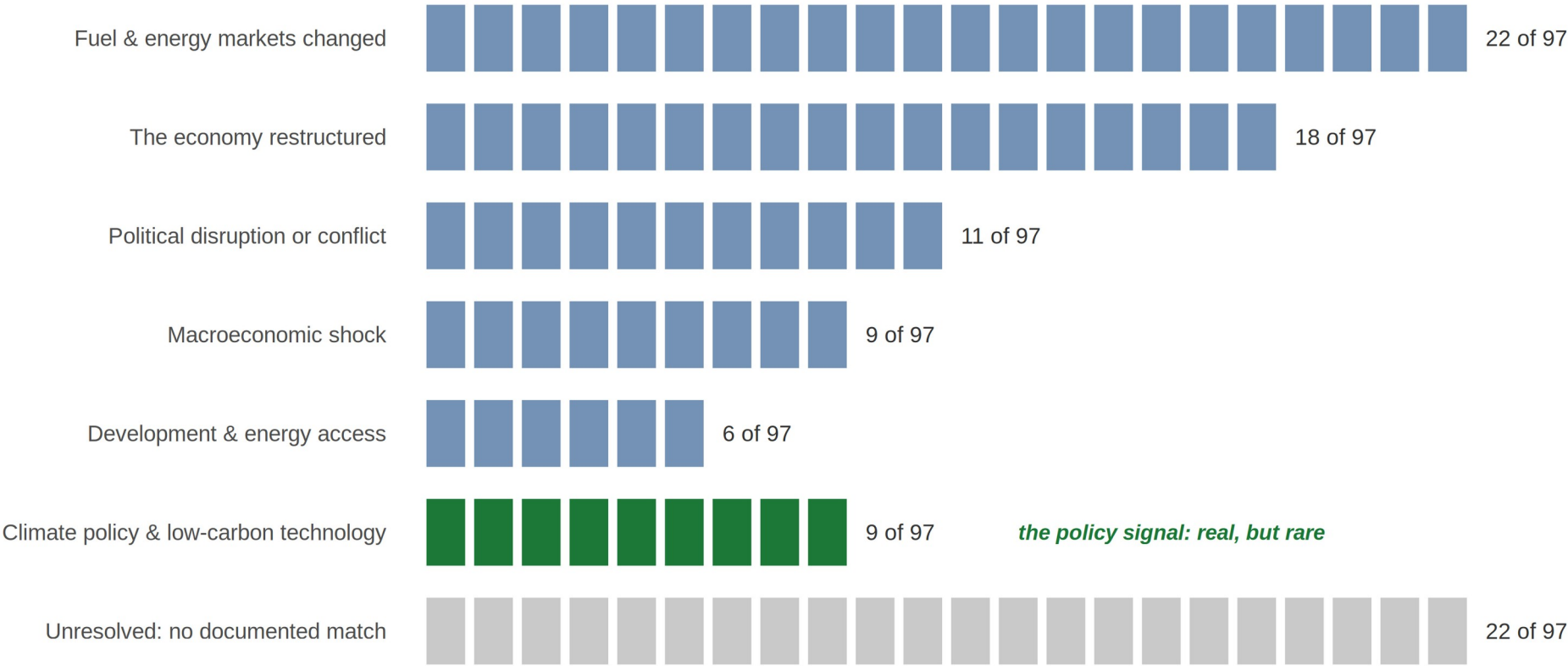


What accompanied the bends

How to read it: One square per country, grouped by the historical process best supported by documentary evidence around its break. Green marks the countries where climate policy and low-carbon technology are that best-supported process; light gray means the record cannot yet explain the break — reported honestly rather than forced into a category.

What accompanied the 97 bends? Mostly markets, restructuring, disruption

One square per country: the best-supported historical mechanism for each statistically selected headline breakpoint



Seven mechanisms, six verdicts

MECHANISMS — what accompanied the bend

Policy-enabled low carbon — deliberate climate/energy policy made physical: infrastructure, fuel mix, market design.

Fuel & energy markets — market-driven fuel switching or supply transformation.

Economic restructuring — industry to services, liberalization, post-socialist transition.

Macroeconomic shock — recessions and financial crises that cut energy use.

Political disruption — wars, state collapse, sanctions, regime change.

Development & energy access — electrification and modernization.

No attributed mechanism — a real, eligible break the record cannot yet explain (22 of 97).

VERDICTS — four tests of the outcome

Did the trend improve? Did it last? Did absolute CO2 actually fall? Was the underlying process constructive?

Constructive persistent decarbonization (9) — passes all four — the gold standard.

Persistent fuel substitution (13) — durable supply-side switch; narrower transformation.

Intensity gain; CO2 still rising (12) — relative gain overwhelmed by growth.

Contractionary / disruptive (20) — the 'improvement' is a crisis in disguise.

Incomplete / nonpersistent (21) — the bend reversed or did not hold.

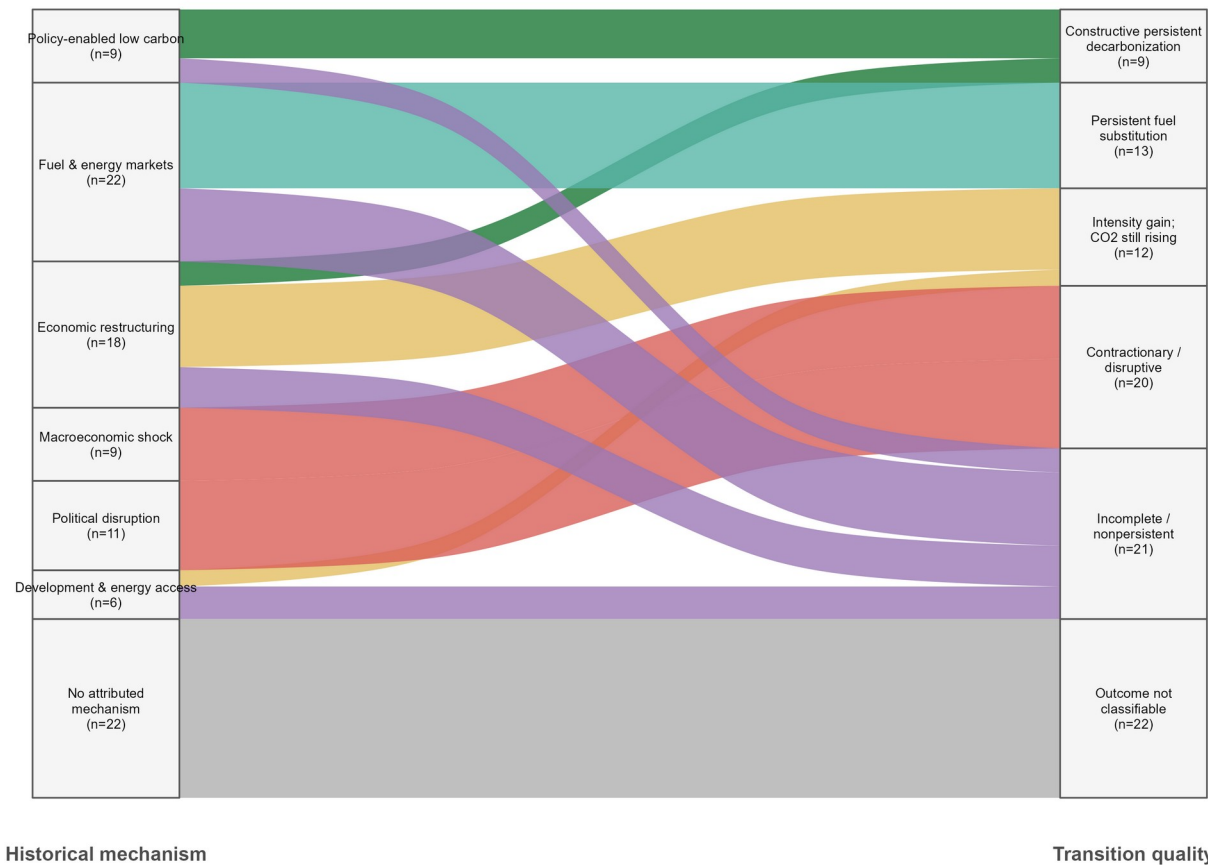
Outcome not classifiable (22) — too little post-break evidence.

From mechanism to verdict: 97 countries, six outcomes

How to read it: Each ribbon carries countries from the process that accompanied their break (left) to how the outcome is judged (right); ribbon width is the number of countries. Follow any left-hand category to see where those transitions actually ended up.

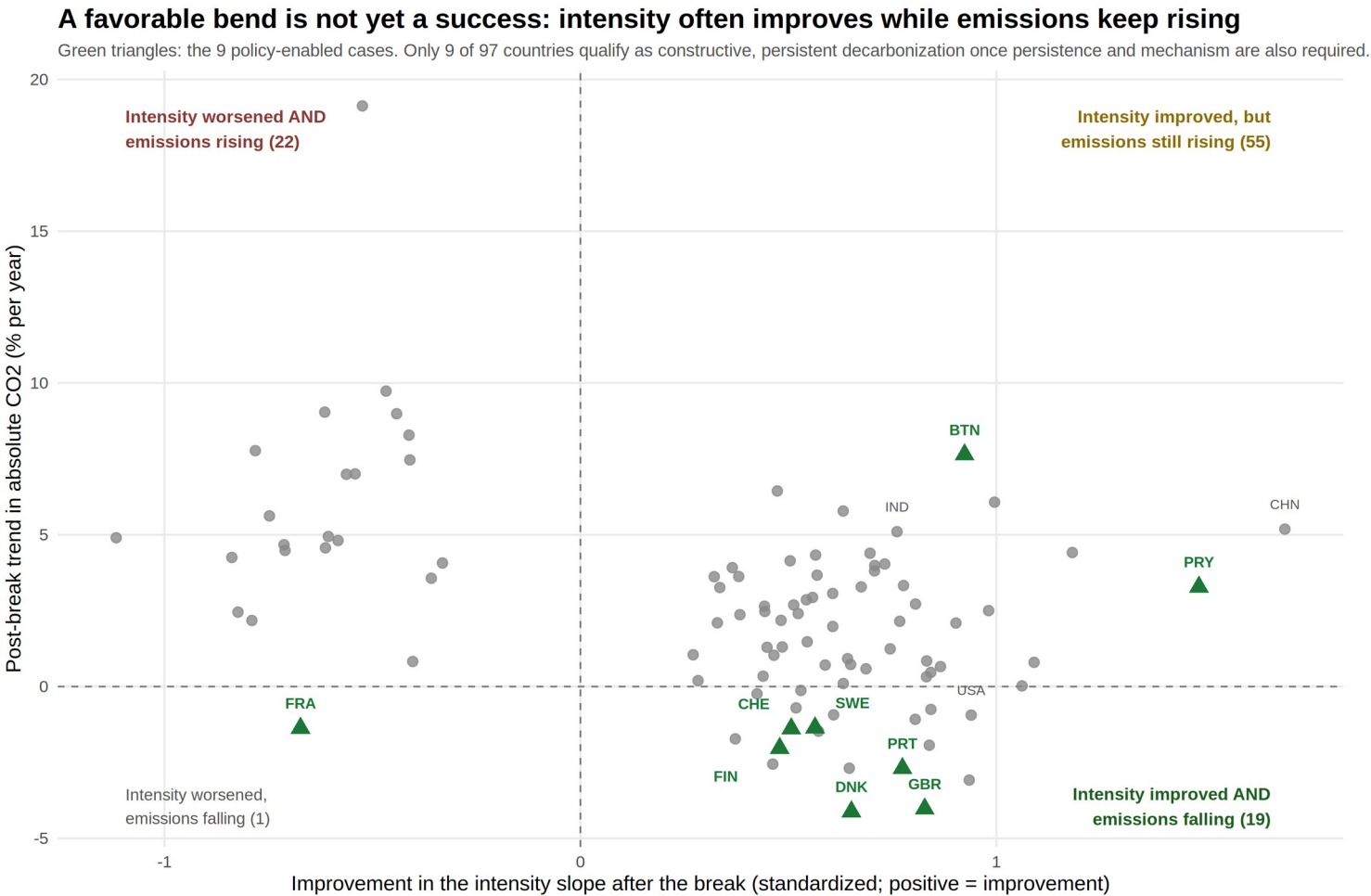
A favorable bend is not automatically a successful transition

Flows of 97 countries from the mechanism accompanying the break to the realized transition-quality classification; residual categories shown at the bottom of each axis



Lower intensity is not the same as falling emissions

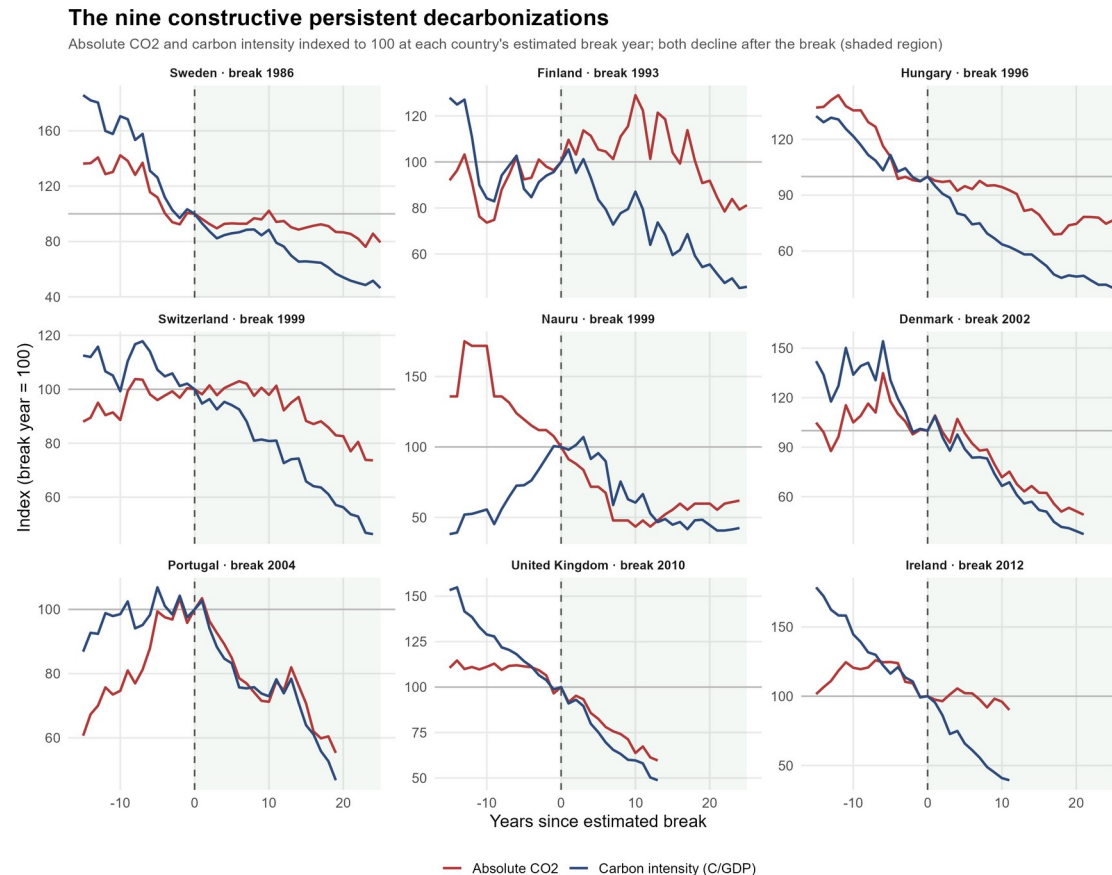
How to read it: Each dot is a country. Further right means its intensity trend improved more after the break; below the dashed horizontal line means absolute emissions are actually falling. Success lives only in the lower right — and most improving countries sit above the line, where growth still outruns their gains.



The nine constructive persistent decarbonizations

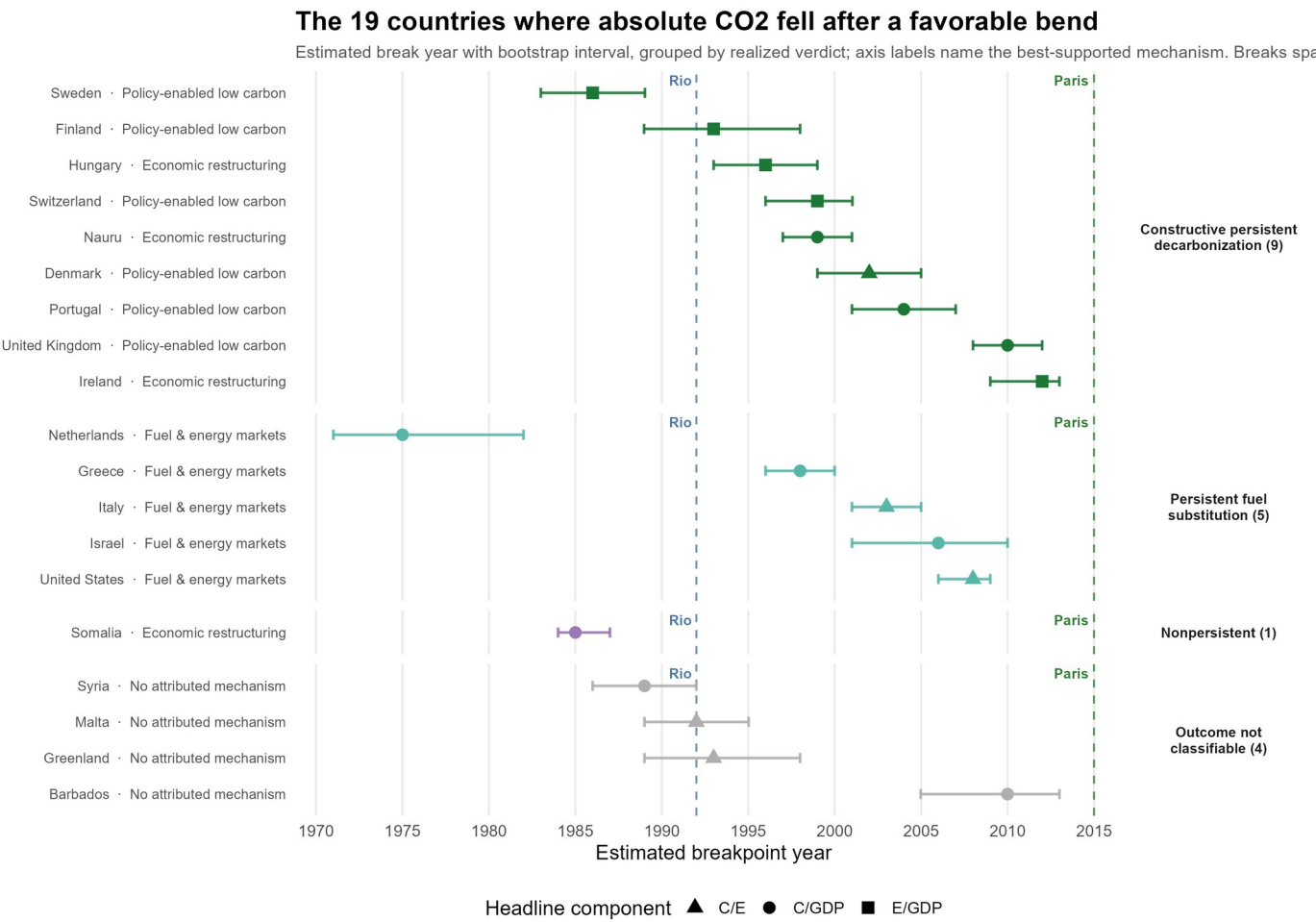
Sweden 1986 • Finland 1993 • Hungary 1996 • Switzerland 1999 • Nauru 1999 • Denmark 2002 • Portugal 2004 • United Kingdom 2010 • Ireland 2012

How to read it: Each panel is one country. Both lines are set to 100 at its break year: blue is carbon intensity, red is absolute CO₂. The defining test is that both fall together after the break (shaded region) — intensity improved, and total emissions genuinely declined.



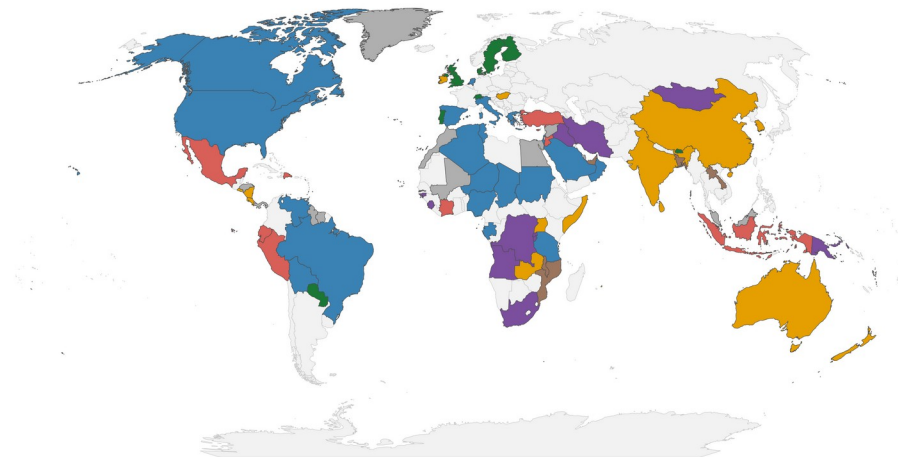
Nineteen countries cut emissions — ten don't earn the verdict

How to read it: Each row is a country where absolute CO2 fell after a favorable bend. The whisker is the uncertainty around its break year; rows are grouped by the verdict, and each label names the best-supported mechanism. Falling emissions alone are not enough — the route matters.

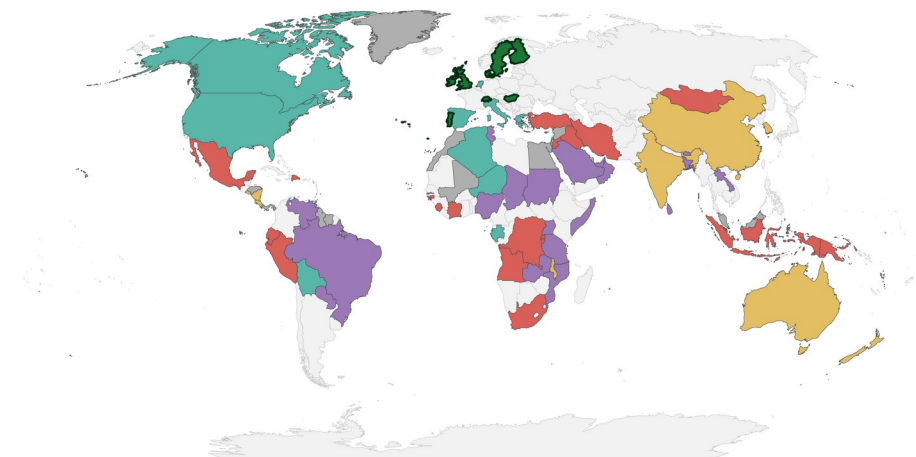


Mechanisms are regional; success is concentrated

How to read it: Left map: each country is colored by the process that accompanied its break. Right map: the same countries colored by outcome verdict — dark green marks the nine constructive cases, concentrated in northern Europe.



Development & energy access Economic restructuring Fuel & energy markets
Macroeconomic shock No attributed mechanism Policy-enabled low carbon
Political disruption



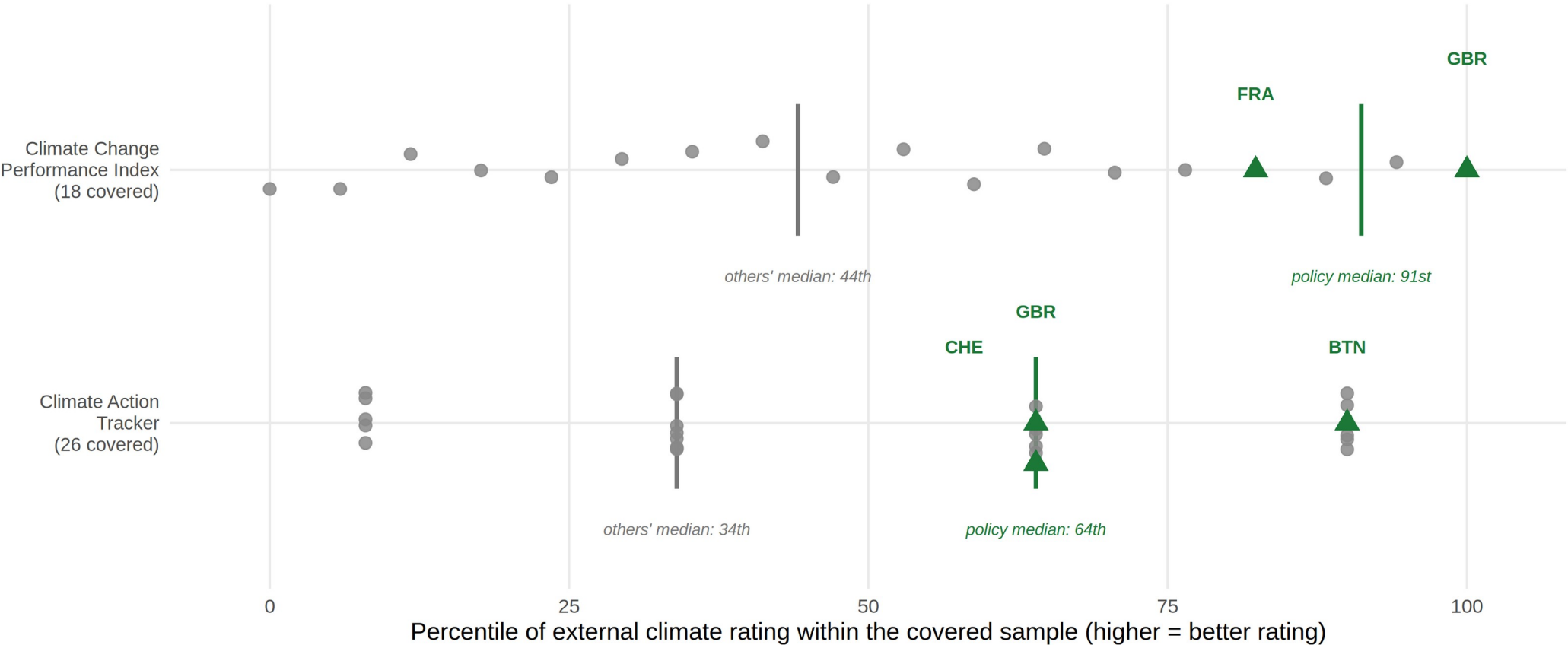
Constructive persistent decarbonization Contractionary / disruptive
Incomplete / nonpersistent Intensity gain; CO2 still rising
Persistent fuel substitution Outcome not classifiable

Independent raters recognize the same countries

How to read it: Each dot is a country covered by the rating, placed by where it ranks among covered countries (right = rated better). Green triangles are our policy-enabled cases; the vertical bars mark each group's median. The policy cases rank high on both ratings — but only 3 and 2 of them are covered, so we show every point rather than hide the small sample.

Independent climate raters rank the policy-enabled cases high, but coverage is thin

External-rating percentile among covered countries. Green triangles: policy-enabled cases (CAT covers 3 of 9; CCPI covers 2). Corroborative, not definitive.



Robustness, corroboration, and honest limits

Consumption-based accounting

When emissions embodied in trade are assigned to consumers instead of producers, only 6 of 17 comparable countries keep a break within 5 years of their production-based date; the median shift is ~12 years. Some 'domestic' transitions partly reflect offshoring — a main result, not a footnote.

Carbon pricing & policy stringency

Carbon pricing is associated with a cleaner fuel mix in panel models with country and year controls; effects on the other components are imprecise. Policy-stringency data cover only 15 of 28 focal countries. Associational, not causal.

Event-timing validation

45% of documented events fall within 2 years of the estimated break versus ~17% if dates were drawn at random; 80% vs 37% within 5 years; 70% vs 33% inside the episode. The historical classifications carry real temporal information — far beyond retrospective storytelling.

What breakpoints can't tell you

Break timing identifies change, not cause. Attribution and outcome classification each leave 22 cases open — restraint that makes the resolved 75% credible. And there is no counterfactual: a treaty that prevents backsliding produces no break at all.

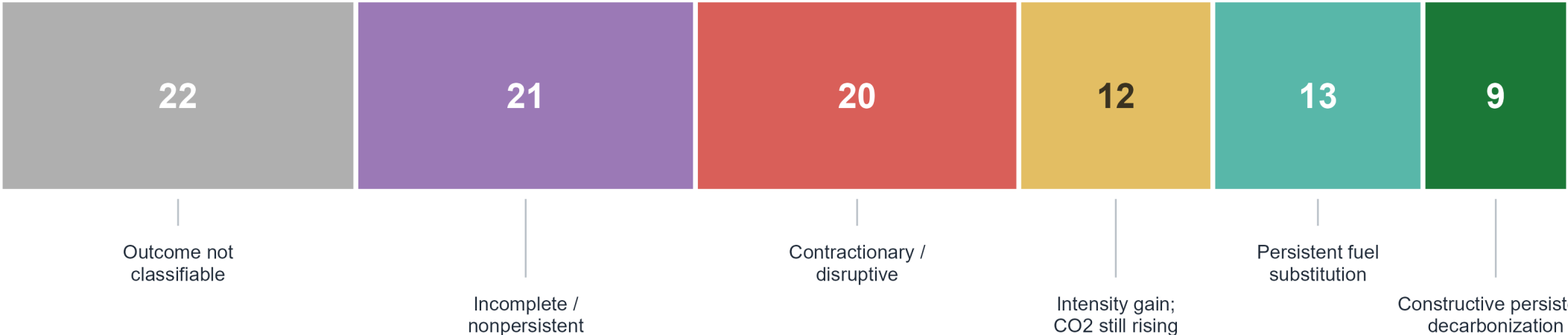
THE VERDICT

Curves bend everywhere; decarbonization is rare

How to read it: All 97 breaks looked statistically favorable. The colors show what they turned out to be once persistence, mechanism, and absolute emissions are checked — only the dark green block on the right is unambiguous decarbonization.

97 statistically eligible national breaks —
every one a real, favorable-looking bend

only 9 pair a persistent intensity decline
with falling absolute CO2



Global agreements can organize climate action — durable national transformations bend the curve.

- Global curves bent in the 1970s; since ~2013 they bend faster — a real supply-side acceleration whose timing cannot yet separate Paris from the renewables revolution it helped consolidate.
- Treaty-adjacent signals exist — especially around Rio — but few survive placebo scrutiny, and national transitions scatter across five decades.
- Favorable bends often reflect recession, disruption, deindustrialization, or offshoring; intensity gains are routinely overwhelmed by growth.
- Nine countries pair persistent intensity declines with falling absolute CO2 — policy made physical in fuel mix, infrastructure, and economic structure.

*Structural breaks say when · Kaya says through which pathway · historical validation says what plausibly happened ·
transition quality says whether it mattered*